

Final Exam - Math 150 - Fall 2025

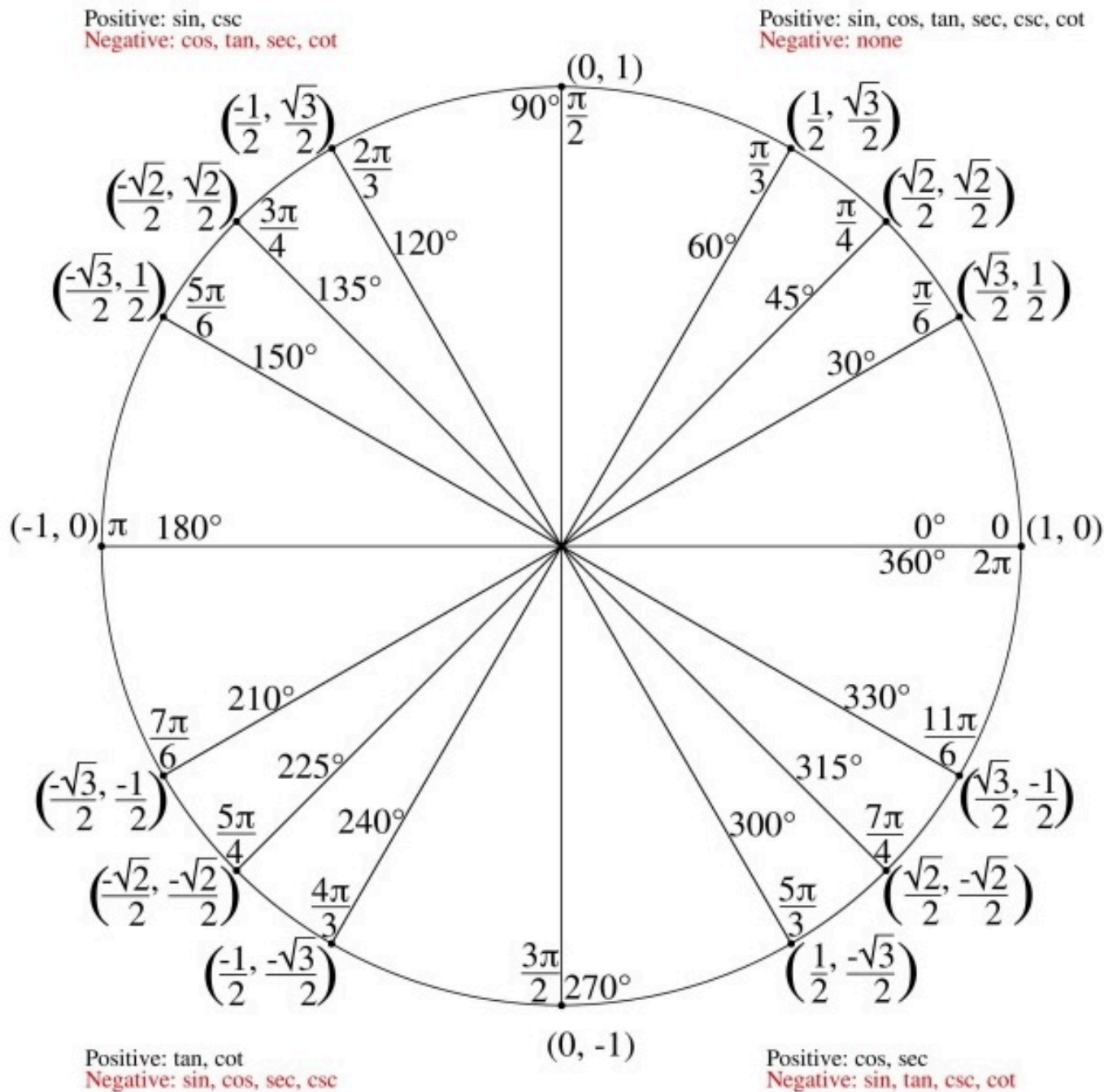
Name _____

Total extra pages _____

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- Please **show your work** for all problems. Use additional pages if necessary.
 - Clearly **label** your work with the problem number you are working on.

Problem number	Point Adjustment	Out of	Notes
1		1	
2		1	
3		1	
4		1	
5		1	
6		1	
7		1	
8		1	
9		1	
10		1	
Total adjustment		10	ALEKS SCORE: Adjusted total score:

Unit Circle & Trigonometric Functions



Trigonometric Function Definitions

$$\begin{aligned}\tan \theta &= \frac{\sin \theta}{\cos \theta} = \frac{y}{x} \\ \cot \theta &= \frac{\cos \theta}{\sin \theta} = \frac{x}{y} \\ \sec \theta &= \frac{1}{\cos \theta} = \frac{r}{x} \\ \csc \theta &= \frac{1}{\sin \theta} = \frac{r}{y}\end{aligned}$$

For each point (x, y) on the unit circle
 $(x, y) = (\cos(\theta), \sin(\theta))$

Important Trigonometric Values

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	0

Derivative Rules

Tips:

- Consider rewriting the function before you compute the derivative.
 - You do **NOT** need to simplify after you compute the derivative.
-

Basic Derivative Rules

$$\frac{d}{dx}[c] = 0 \quad (\text{Constant Rule})$$

$$\frac{d}{dx}[x] = 1$$

$$\frac{d}{dx}[cf(x)] = c \frac{d}{dx}[f(x)] \quad (\text{Constant Multiple})$$

(Sum Rule)

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$$

$$\frac{d}{dx}[x^n] = nx^{n-1} \quad (\text{Power Rule})$$

(Product Rule)

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + g'(x)f(x)$$

(Quotient Rule)

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - g'(x)f(x)}{[g(x)]^2}$$

(Chain Rule)

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

Other tips

$$\frac{1}{x^n} = x^{-n}$$

$$x^{\frac{m}{n}} = \sqrt[n]{x^m}$$

Point-Slope form of a line
with slope m and point (x_1, y_1) :

$$y = m(x - x_1) + y_1$$

Derivatives of log and exponential functions

$$\frac{d}{dx}[e^x] = e^x$$

$$\frac{d}{dx}[a^x] = a^x \ln(a)$$

$$\frac{d}{dx}[\ln(x)] = \frac{1}{x}, \quad x > 0$$

$$\frac{d}{dx}[\log_a(x)] = \frac{1}{x \ln(a)}$$

Derivatives of trig functions

$$\frac{d}{dx}[\sin(x)] = \cos(x)$$

$$\frac{d}{dx}[\cos(x)] = -\sin(x)$$

$$\frac{d}{dx}[\tan(x)] = \sec^2(x)$$

$$\frac{d}{dx}[\cot(x)] = -\csc^2(x)$$

$$\frac{d}{dx}[\sec(x)] = \sec(x) \tan(x)$$

$$\frac{d}{dx}[\csc(x)] = -\csc(x) \cot(x)$$

Basic Differentiation Rules	Basic Integration Rules
$\frac{d}{dx}C = 0$	$\int 0 \, dx = C$
$\frac{d}{dx}(kx) = k$	$\int k \, dx = kx + C$
$\frac{d}{dx}[kf(x)] = kf'(x)$	$\int [kf(x)] \, dx = k \int f(x) \, dx$
$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$	$\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx \pm \int g(x) \, dx$
$\frac{d}{dx}x^n = nx^{n-1}$	$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \, n \neq -1 \quad (\text{Power Rule})$
$\frac{d}{dx} \sin x = \cos x$	$\int \cos x \, dx = \sin x + C$
$\frac{d}{dx} \cos x = -\sin x$	$\int \sin x \, dx = -\cos x + C$
$\frac{d}{dx} \tan x = \sec^2 x$	$\int \sec^2 x \, dx = \tan x + C$
$\frac{d}{dx} \cot x = -\csc^2 x$	$\int \csc^2 x \, dx = -\cot x + C$
$\frac{d}{dx} \sec x = \sec x \tan x$	$\int \sec x \tan x \, dx = \sec x + C$
$\frac{d}{dx} \csc x = -\csc x \cot x$	$\int \csc x \cot x \, dx = -\csc x + C$
$\frac{d}{dx} e^x = e^x$	$\int e^x \, dx = e^x + C$
$\frac{d}{dx} \ln x = \frac{1}{x}, \, x > 0$	$\int \frac{1}{x} \, dx = \ln x + C$

Other tips:

- A function is concave up on the interval (a, b) if $f''(x) > 0$ for all x in (a, b) .
- A function is concave down on the interval (a, b) if $f''(x) < 0$ for all x in (a, b) .
- If $f(x)$ is a continuous function on the interval $[a, b]$ and $F'(x) = f(x)$, then $\int_a^b f(x) \, dx = F(b) - F(a)$.
- The point-slope equation of a line with slope m and through the point (x_1, y_1) is $y = m(x - x_1) + y_1$.
- $\frac{1}{x^n} = x^{-n}$ and $x^{1/m} = \sqrt[m]{x}$.
- $(x^2 - a^2) = (x - a)(x + a)$
- $(x^3 - a^3) = (x - a)(x^2 + ax + a^2)$
- $(x + a)^3 = x^3 + 3ax^2 + 3a^2x + a^3$

Student Name: _____

Class Name: **Calculus 1 - fall 2025**

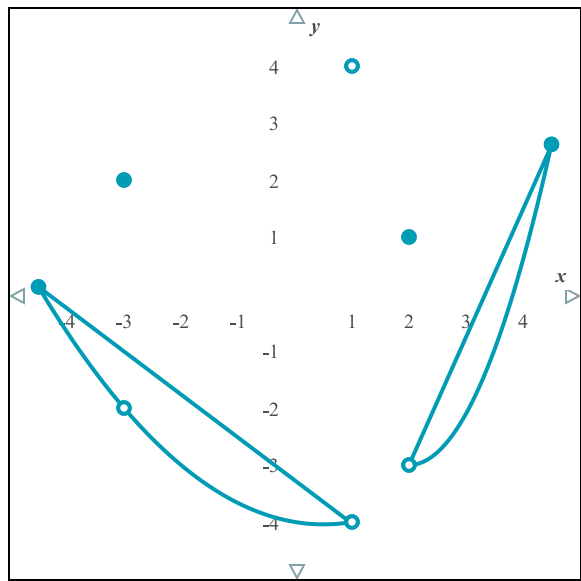
Number of Questions: **10**

Instructor Name: 

Instructor Note: **DLDRC**

Question 1 of 10

The function h is graphed below.



Find the following limits. If a limit does not exist, write "Does Not Exist".

$$\lim_{x \rightarrow -2^-} h(x) =$$

$$\lim_{x \rightarrow -2^+} h(x) =$$

$$\lim_{x \rightarrow -2} h(x) =$$

Question 2 of 10

Find the following limit.

If the limit does not exist, write "Does Not Exist."

$$\lim_{x \rightarrow -1} \frac{2x^2 + 3x - 7}{3x + 2}$$

Question 3 of 10

Find the following limit.

If the limit does not exist, write "Does Not Exist."

$$\lim_{x \rightarrow 0} \frac{x^4 + 2x^2 - x}{2x}$$

Question 4 of 10

Let $f(t) = 3t^2 + 5t$.

(a) For $h \neq 0$, find the difference quotient $\frac{f(7+h) - f(7)}{h}$.

Simplify your answer so that the resulting expression is defined at $h = 0$.

$$\frac{f(7+h) - f(7)}{h} = \underline{\hspace{2cm}}$$

(b) Find $f'(7)$.

$$f'(7) = \underline{\hspace{2cm}}$$

Question 5 of 10

Find the derivative of the function.

$$g(z) = \frac{2z^3}{1+z^2-z^4}$$

$$g'(z) =$$

Question 6 of 10

Find an equation of the tangent line to the curve $y = (x+4)(3x+1)$ at the point $(-1, -6)$.

Question 7 of 10

Find all values of x at which the curve has a horizontal tangent line.

$$y = -2x^3 - 8x^2 + 6x + 11$$

Write your answers as exact values (not decimal approximations). If there is more than one such value, separate them with commas. If there is no such value, click on None.

$$x =$$

Question 8 of 10

Let the function f be defined as follows.

$$f(x) = \frac{1}{2}x^3 + 9x^2 + 7x - 3$$

- (a) Write your answers as an interval or list of intervals. When writing a list of intervals, make sure to separate them with commas and to use as few intervals as possible. Write "None" if applicable.

Write each answer as an interval or a list of intervals. When writing a list of intervals, make sure to separate them with commas and to use as few intervals as possible. Click on "None" if applicable.

Interval(s) on which f is concave up: _____

Interval(s) on which f is concave down: _____

- (b) Find all the inflection points (points of inflection) of the function f . Write the points as ordered pairs. Use exact values in your answers (not decimal approximations).

If there is more than one point, separate them with commas. Click on "None" if applicable.

All inflection points of f : _____

Question 9 of 10

Consider the function defined by the equation below.

$$y = 3\sqrt{x} - 3x$$

Find the absolute minimum and absolute maximum values of y on the interval $[1, 9]$. Give exact answers, not decimal approximations. If there is no absolute minimum or absolute maximum value, write "None".

Absolute minimum value: _____

Absolute maximum value: _____

Question 10 of 10

Evaluate the integral. Give an exact answer and not a decimal approximation.

$$\int_{-1}^2 (5z^3 + 3z - 8) dz = \underline{\hspace{2cm}}$$